



EntertAlnment

An Experiment With Machine Learning and It's Ability to Write
Cohesive and Original Short Stories/Films, and It's Effect on
the Entertainment Industry

LIAM COLL



Contents

Introduction	3
The Experiment.....	3
The Tests	5
Findings.....	7
Analysis	7
Conclusion	9
Bibliography	10

This is the pdf version of a research project intended to be presented as a web-article. Therefore it is best viewed on its web-page at <http://liamcoll.com/AI.html>, where along with all content featured in this document, there is the text output from the machine-learning text generator as well as embedded content.

Introduction

Artificial Intelligence technologies are commonly treated with great deals of fear and concern when they are reported on in the news, and when they appear in films and television. These fears mostly consist of either AIs becoming self-aware, or AIs automating so much of human society that there will be no more jobs left for humans. (Fast & Horvitz, 2016)

Many of these fears are created, ironically, from science fiction stories and create unrealistic perspectives on AI (you can see a video essay I created on this very topic here. (Coll, 2020)

I decided to investigate AI science fiction, and conduct a meta-experiment:

Can an AI write a short story akin to the ones that contributed so much to its own historical development and treatment?

Could the story it generates become a replacement for a “real”, human written story?

Could the AI, in a sense, replace human jobs by creating stories about itself, or a system of a similar nature, taking over human jobs?

The Experiment

In order to conduct this experiment, I took five short stories about Artificial Intelligence from different periods of time, and ran them through Textgenrnn, a Python module created by Max Woolf. The scripts that Max had written were on Google Colaboratory, a service that allows you to run Python scripts in a browser, with access to Google GPUs. Basically, I was able to run text-generation machine learning quickly, in just my browser.

Max's scripts allow you to input text, tweak a few variables and train text-generation models.

The variables I altered during the experiment were:

- The number of epochs/cycles the model trains for
- The temperatures of the model after it has been trained. Temperatures is a number between 0 and 1 which allows you to decide how “unpredictable” the generated text will be. The provided scripts allows for the generator to cycle through 5 different temperatures as it generates, for example [1.0, 0.2, 0.3, 0.6, 0.9]
- Whether the model is word or character level. Basically determines whether the model learns the patterns of the words, or the characters.

I used the following texts:

- Patterns of a Murmuration, in Billions of Data Points - JY Yang (Yang, 2014)
- I Have No Mouth and I must Scream - Harlan Ellison (Ellison, 1977)
- The Machine Stops - E. M. Forster (Forster, 1909)
- R.U.R. (Rossum's Universal Robots) - Karel Čapek (Capek et al., 1920)
- The Last Question - Isaac Asimov (Asimov, 1956)

These texts were chosen for their significance to the science fiction genre, and particularly the theme of Artificial Intelligence.

Each text was, on average, 33097 characters long. I decided that the model should generate a short-story that when adapted into a short film, would last for 5 minutes, which is, again on average, 5 pages, or 15000 characters. After deciding on these parameters, I trained 4 different models and generated a few text files from each.

The Tests

Test 1:

Level: Character

Epochs: 700

Temperature: 1.0, 0.5, 0.2, 0.2

This test ultimately was a failure. The varying temperatures caused the words to make sense structurally, but the model failed to generate real words in the English language.

However, the text is still entertaining in a sense, there's a certain humour to watching a computer attempt to emulate human speech.

Test 2:

Level: Character

Epochs: 700

Temperature: 1.0, 1.0, 1.0, 1.0

I decided to remedy the nonsensical words by adjusting the temperatures of the text generator, setting them all to expected tokens. This resulted in real English words. I was happy with the result and even started to convert the story into a screenplay.

While undergoing this process I realised that the text was lifted directly from I Have No Mouth and I Must Scream. The characters had become so expected that it was just rewriting the input text.

Test 3:

Level: Word

Epochs: 700

Temperature: 1.0, 1.0, 1.0, 1.0

The word level generator gave sensible words, but struggled when it came to legibility. The sentences weren't formed correctly, and there were a lot of repeated words.

There were also lots of randomly placed commas. This phenomenon occurred with all of the text files I generated.

Test 4:
Level: Character
Epochs: 700
Temperature: [0.1, 0.3, 0.2, 0.2]

This final test just generated the text from “The Machine Stops”, and at the end begins to repeat on itself.

Findings

In all the models that I tried, none came even close to generating original, sensical text, let alone anything with a coherent plot or original characters. I experimented with different kinds of models and adjusted parameters, but didn't manage to generate anything remotely similar to what a human could write. While using temperatures of 0.8-1.0, sensical text was generated, but was often lifted directly from the input text.

When using lower temperatures, nonsensical words were created.

Analysis

The findings from this experiment can provide insight into the threat of AI against the entertainment industry, particularly in the writing industry. While reading the output text files, it is apparent that text generation machine learning technology is not an immediate problem that will have to be addressed.

There are, of course, limitations to the method I used to test the capabilities of text generation machine learning.

Textgenrnn has shortcomings, as outlined by the creator Max Woolf:

"You will not get quality generated text 100% of the time, even with a heavily-trained neural network. That's the primary reason viral blog posts/Twitter tweets utilizing NN text generation often generate lots of texts and curate/edit the best ones afterward."

"Results will vary greatly between datasets. Because the pretrained neural network is relatively small, it cannot store as much data as RNNs typically flaunted in blog posts. For best results, use a dataset with at least 2,000-5,000 documents." (Woolf, 2017/2020)

In order to create a story with particular traits: in this case being 15000 characters long, being about AIs/Al takeovers, being in a science-fiction style of writing, 2000-5000 texts with these traits must be sourced and a model trained on this dataset. This is difficult, if 2000-5000 of these texts even exist at all.

An alternative is to create 'genre models', which generate stories in a particular genre, such as science-fiction.

However, as well as running into the common pitfalls of AI generated text such as legibility, poor structure and nonsensical narratives, even more narrative control is lost. Genres deal with many different sub-genres and various themes and topics which cannot be controlled with trained models.

Another way to improve the process is to write text in the style of AI Dungeon (Latitude, n.d.), where the user can influence the text generation as it generates, adding prompts, actions, and dialogue to push the narrative where they want it to go. This still requires human input but is a lot more 'hands-off'.

Consideration must be made to the chosen method of text generation during the experiments. Textgenrnn was chosen due to its ease of use and accessibility, however is far from the most powerful text generator available.

Weaknesses in Textgenrnn lie in the fact that it is a "model free" text generator. It

doesn't understand what a word is, what a word consists of, and how it should be combined with other words to create a sensible output.

Legibility can be increased by using model-based text generation, and using a lexical English database such as WordNet (Jing, 1998).

OpenAI's GPT-2 is a powerful model trained on 40GB of text from the internet. It has been used to generate [poetry](#) (Piper, 2019), [novels](#) (Samuel, 2019) and even candy heart messages.

For entertaining and compelling narratives to be written, the most powerful models available at the time must be used.

In any regard, the text generated by GPT-2 is still strange. In most cases, it uses unusual words *that just don't feel right* – an uncanny-valley of words. How often would you find a candy-heart that just says “ALL HAIL THE CHICKEN”?

This makes AI-generated text a poor replacement for human-written text, but entertaining in itself. AI creates text so unusual and random that it becomes humorous, which some people have acted upon, for example the previously-mentioned candy-hearts, the AI-generated Harry Potter excerpt [Harry Potter and What Looked Like a Large Pile of Ash](#) (Botnik Studios, 2018), and [I taught an AI to make pasta](#) (Cruz, 2020).

Conclusion

All of this analysis leads to the conclusion that AI is not a threat to the entertainment and writing industries. AI is unable to generate legible text most of the time, and when it does it struggles with narrative consistency, if a narrative exists at all.

This makes AI an unfit replacement for human writers. Even if an AI managed to create a story with a coherent plot, it wouldn't be able to create compelling original characters, as well as unique plots. AI text generation has potential to be entertaining in itself, almost as its own sub-genre, but is unlikely to replace humans in the industry anytime soon.

AI Text generation is entertainment in itself, but not replacement for serious writing

Difficult to create plots with particular plot points and characters

Therefore not a threat

Bibliography

Asimov, I. (1956). *The Last Question*.

Botnik Studios. (2018). *Harry Potter and What Looked Like a Large Pile of Ash*.

<https://botnik.org/content/harry-potter.html>

Capek, K., Novack-Jones, C., & Klima, I. (1920). *R.U.R. (Rossum's Universal Robots)*. Penguin Publishing Group. <https://books.google.co.nz/books?id=-P-TiB7CzGkC>

Coll, L. (2020, August 28). *The Influence of Sci-Fi AIs on Real Life AIs and AI Media Reporting*. <https://www.youtube.com/watch?v=373rv4iLo1c&feature=youtu.be>

Cruz, S. (2020, May 27). *I taught an AI to make pasta*.

https://www.youtube.com/watch?v=Y_NvR5dlaOY

Ellison, Harlan. (1977). *I have no mouth and I must scream: Stories*. Pyramid Publications; /z-wcorg/.

Fast, E., & Horvitz, E. (2016). *Long-Term Trends in the Public Perception of Artificial Intelligence*. <https://arxiv.org/pdf/1609.04904.pdf>

Forster, E. M. (1909). *The Machine Stops*. *Lits*.

<https://books.google.co.nz/books?id=siYDkgAACAAJ>

Jing, H. (1998). *Usage of WordNet in Natural Language Generation*. 7.

Latitude. (n.d.). *AI Dungeon*. Retrieved October 19, 2020, from

<https://play.aidungeon.io/main/home>

Piper, K. (2019, May 15). *A poetry-writing AI has just been unveiled. It's ... Pretty good*. Vox. <https://www.vox.com/2019/5/15/18623134/openai-language-ai-gpt2-poetry-try-it>

Samuel, S. (2019, August 30). *How I'm using AI to write my next novel*. Vox.

<https://www.vox.com/future-perfect/2019/8/30/20840194/ai-art-fiction-writing-language-gpt-2>

Woolf, M. (2020). *Minimaxir/textgenrnn* [Python].

<https://github.com/minimaxir/textgenrnn> (Original work published 2017)

Yang, J. (2014). *Patterns of a Murmuration, in Billions of Data Points*.